

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 Fill in the following blanks with the correct choice of the words from this list:

Increasing, decreasing, positive, negative, concave up, concave down

- (a) If you know $f''(x) > 0$, then you know $f'(x)$ is _____ and $f(x)$ is _____.

Explanation. increasing, concave up

- (b) If you know $g'(x) < 0$ and decreasing, then you know $g(x)$ is _____ and _____.

Explanation. decreasing, concave down

- (c) If you know $h(x)$ is positive, increasing, and concave down, then you know $h'(x)$ is _____ and _____ and that $h''(x)$ is _____.

Explanation. positive, decreasing, negative

[4]

Problem 2

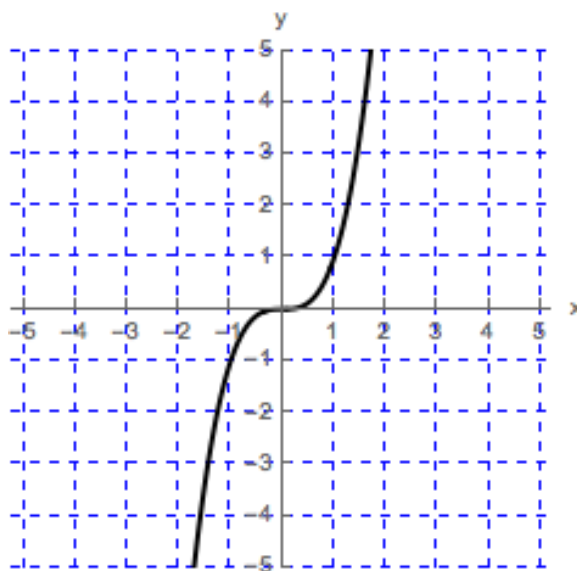
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Problem 2

Problem 2 Sketch a graph of a function that is continuous on $(-\infty, \infty)$ that has the following properties.

- (a) Function f does not have a local maximum or minimum. f contains a point where $f'(x) = 0$

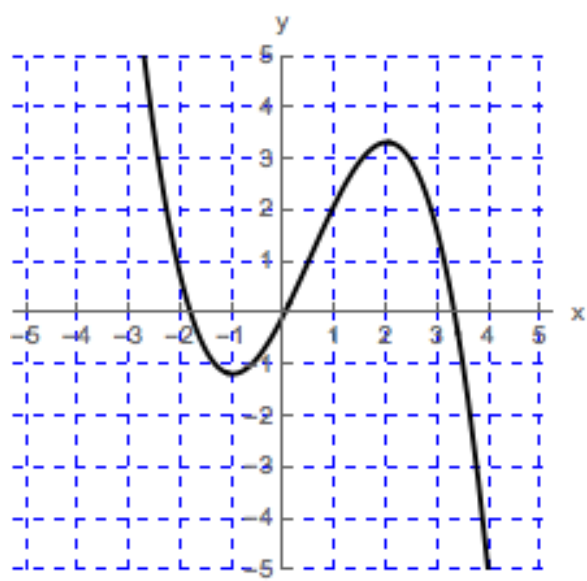
Explanation.



- (b) $g'(x) < 0$ on $(-\infty, -1)$; $g'(x) > 0$ on $(-1, 2)$; $g'(x) < 0$ on $(2, \infty)$.

Explanation.

Problem 2



Problem 3

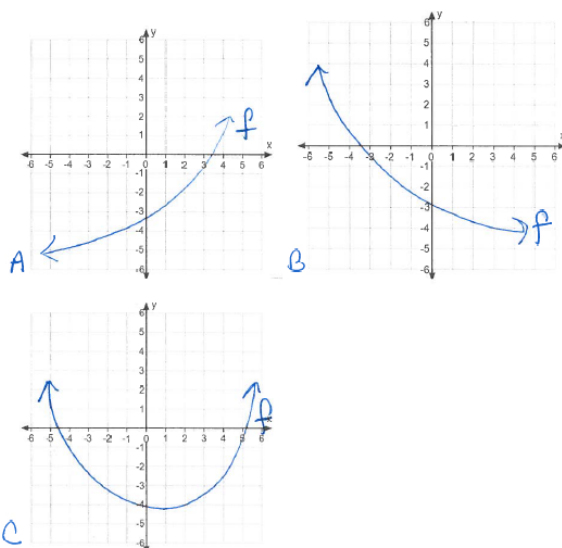
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Problem 3

Problem 3 Give an example or sketch of a function that is continuous on $(-\infty, \infty)$ and satisfies given properties. If such a function does not exist, explain why.

- (a) A function f is concave up and negative everywhere.

Explanation. This is not possible. If a function is always concave up then at some point, the function must cross the x -axis and become positive. See the three example figures below.



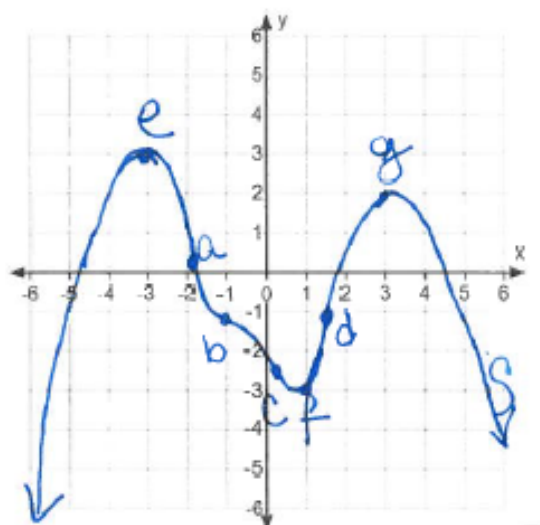
- (b) A function f is decreasing and concave up everywhere.

Explanation. This is possible. See figure B above.

- (c) A function s has exactly 3 local extrema and four inflection points.

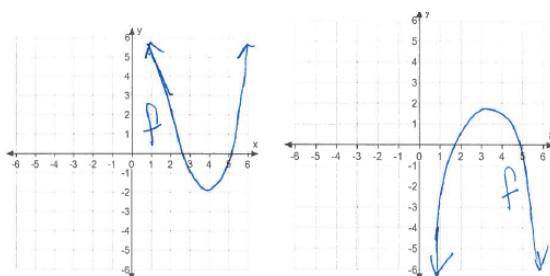
Explanation. This is possible. See the figure below. The graph of s has inflection points at a, b, c, d and local extrema at e, f, g .

Problem 3



(d) A function f has exactly 2 zeros and one local extrema.

Explanation. This is possible. See the figures below for examples.



Problem 4

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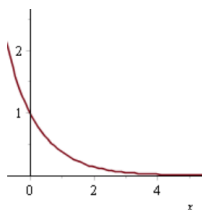
Problem 4

(a) You are given that $f''(x) > 0$ for all x . Which of the following must be true about $f(x)$ on the region $0 \leq x \leq 2$?

- (i) There is a critical point between 0 and 2.
- (ii) There is a local maximum, but not enough information is given to count.
- (iii) f

Expla

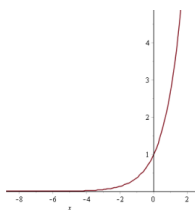
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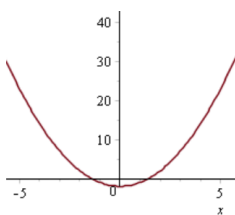
(b) You are told that $f''(x) > 0$ for all x . Which of the following must be true about the graph of $y = f(x)$?

- (i) The graph is a straight line.
- (ii) The graph crosses the x -axis at most once.
- (iii) The graph is concave down.
- (iv) The graph crosses the y -axis more than once.
- (v) The graph is concave up.

Explanation. Only (v) must be true. For (i), the function $f(x) = e^x$ provides a counterexample:



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Part (iii) is clearly false since $f''(x) > 0$ means that f is concave up. Part (iv) is false for any function.

Problem 5

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Problem 5 Suppose a function f satisfies the following conditions:

- (a) $f(0) = 0$ and $f'(-4) = f'(2) = f'(10) = 0$
 - (b) $\lim_{x \rightarrow 6} f(x) = -\infty$, and $\lim_{x \rightarrow +\infty} f(x) = 6$
 - (c) $f'(x) < 0$ on $(-\infty, -4)$, $(2, 6)$, and $(10, +\infty)$
 - (d) $f'(x) > 0$ on $(-4, 2)$, and $(6, 10)$
 - (e) $f''(x) > 0$ on $(-\infty, 0)$, and $(14, +\infty)$
 - (f) $f''(x) < 0$ on $(0, 6)$, and $(6, 14)$
- (a) List the **interval(s)** where the function f is **both increasing and concave UP**.

Explanation. ($-4, 0$)

- (b) List the **interval(s)** where the function f is **both increasing and concave DOWN**.

Explanation. ($0, 2$), $(6, 10)$

- (c) List the **interval(s)** where the function f is **both decreasing and concave UP**.

Explanation. ($-\infty, -4$), $(14, +\infty)$

- (d) List the **interval(s)** where the function f is **both decreasing and concave DOWN**.

Explanation. ($2, 6$), $(10, 14)$

Problem 5

- (e) List the **x-coordinates** at which f has a **local minimum**. Write "none" if appropriate.

Explanation. $x = -4$

- (f) List the **x-coordinates** at which f has a **local maximum**. Write "none" if appropriate.

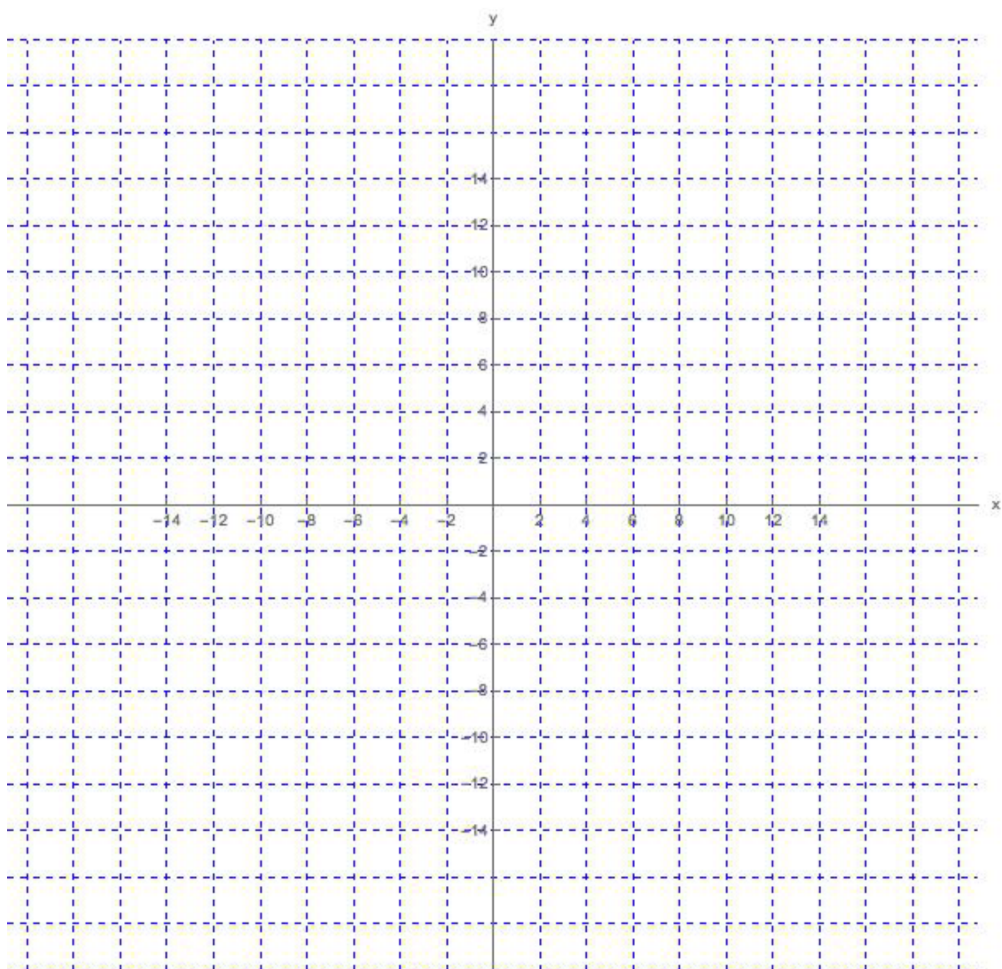
Explanation. $x = 2, \quad x = 10$
(f can also have a local maximum at $x = 6$, if we include $x = 6$ in the domain)

- (g) List the **x-coordinates** of all **inflection points** of f . Write "none" if appropriate.

Explanation. $x = 0, \quad x = 14$

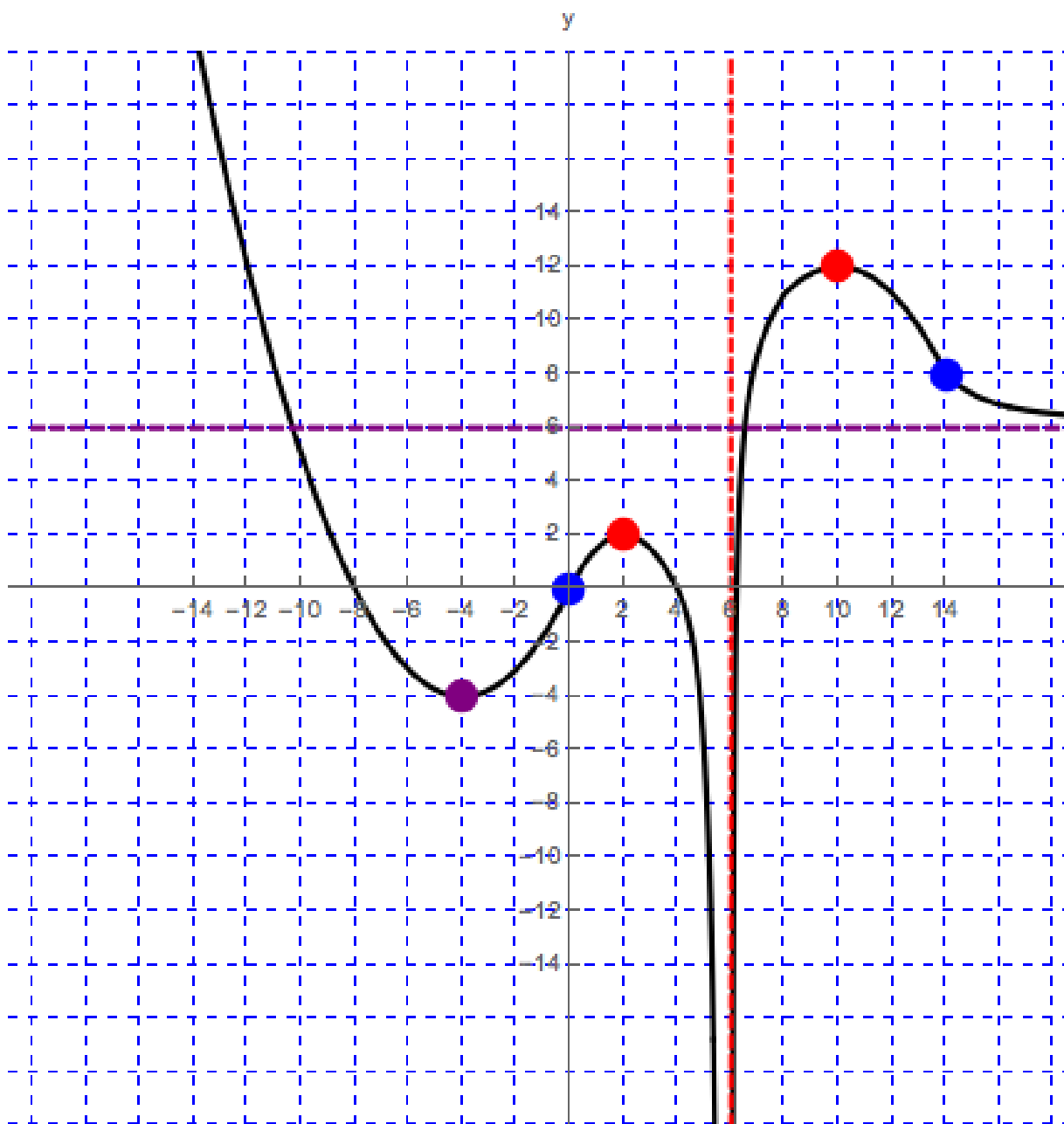
- (h) Sketch the graph of f .

Problem 5



Explanation. This is just one of many possible graphs. The domain of f could be $(-\infty, +\infty)$, and f could satisfy the given conditions.

Problem 5



Purple point : local minimum

Red points : local maxima

Problem 5

Blue points : inflection points

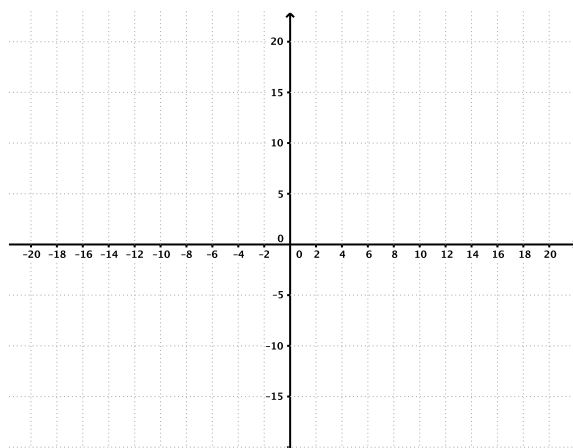
Problem 6

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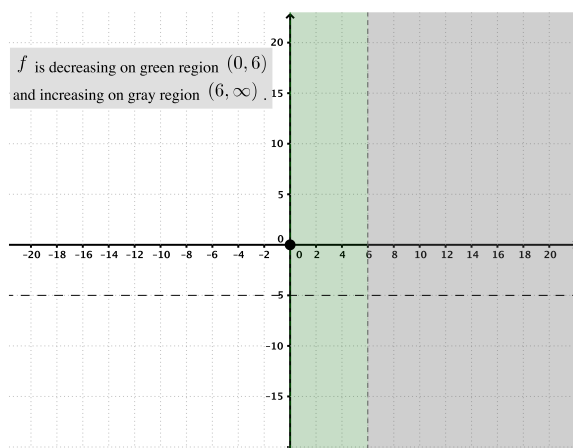
Problem 6

Problem 6 Sketch the graph of a function f satisfying all of the conditions:

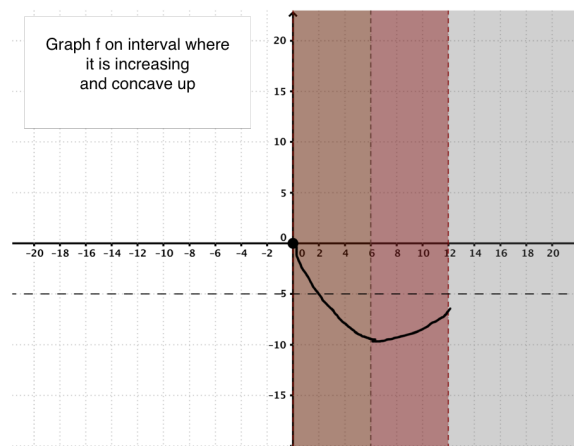
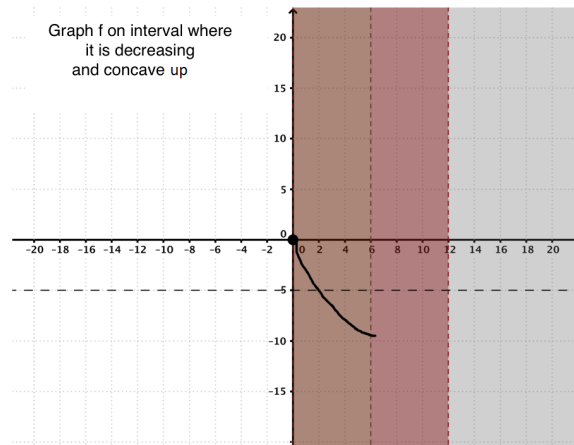
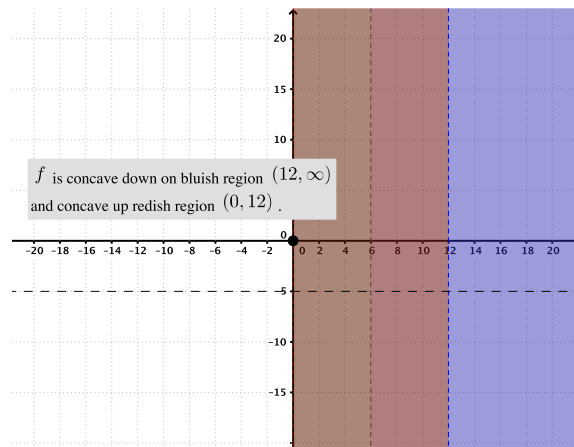
- (a) f is continuous and odd, $f(0) = 0$,
- (b) $\lim_{x \rightarrow \infty} f(x) = -5$,
- (c) $f'(x) > 0$ on $(6, \infty)$,
- (d) $f'(x) < 0$ on $(0, 6)$,
- (e) $f''(x) > 0$ on $(0, 12)$, and
- (f) $f''(x) < 0$ on $(12, \infty)$.



Explanation.



Problem 6



Problem 6

